IoT Communication & Data Pipeline

Mission 2 Project Report

1. Objective

Build an MQTT-based IoT pipeline that transfers simulated sensor readings through a broker to a subscriber and stores the readings in SQLite.

2. Technologies

Python, Paho MQTT, MQTT, Mosquitto/public MQTT broker, JSON, SQLite, CSV and Matplotlib.

3. Architecture

The sensor simulator creates temperature and humidity readings. The publisher sends JSON data to the MQTT topic vertexmind/iot/sensor. The broker routes messages to subscribers. The subscriber validates the payload and stores it in SQLite.

4. MQTT Communication

MQTT uses a publish/subscribe model. The publisher and subscriber communicate through the broker rather than directly. QoS 1 is used in the sample implementation.

5. Data Storage

SQLite provides lightweight persistent storage for device ID, timestamp, temperature, humidity and receive time.

6. Testing

A 30-record sample dataset was generated. An offline local pipeline demo is also included to demonstrate Publish → Subscribe → Store without an external broker.

7. Challenges

The main challenges were understanding MQTT topics, coordinating publisher/subscriber execution, parsing JSON payloads and committing received data to storage.

8. Result

The project demonstrates sensor data generation, MQTT transmission, subscription and database storage as a complete IoT data pipeline.

Sample Data

Time	Temperature °C	Humidity %
2026-09-27 10:00:00	27.84	50.50
2026-09-27 10:01:00	25.65	54.46
2026-09-27 10:02:00	28.42	63.53
2026-09-27 10:03:00	29.35	51.74
2026-09-27 10:04:00	26.53	50.60
2026-09-27 10:05:00	25.31	60.11

2026-09-27 10:06:00	24.16	53.98
2026-09-27 10:07:00	27.90	60.90

Data flow: Sensor → Publisher → MQTT Broker → Subscriber → SQLite → Analysis